Reviewer 1 — Round 2

Recommendation: Major Revision

Does the title reflect the content, specific and effective?

Yes

Further Suggestions:

- The revised title is specific and accurately identifies the hybrid ARIMA-LSTM approach, Indonesian coal-sector stocks, and the forecasting focus.
- The title appropriately avoids claiming universal investment-prediction capability.
- Keep the dashboard as a secondary implementation component, not the principal scientific contribution.

Abstracts and keywords have described the overall content?

Yes

Further Suggestions:

- The abstract is approximately 184 words and complies with the journal template requirement of 100–250 words.
- It clearly states the four stocks, data period, out-of-sample test period, baseline comparison, metrics, stock-specific results, and exploratory nature of long-term projections.
- Clarify the forecasting horizon used in the evaluation: state explicitly whether forecasts are one-step-ahead rolling forecasts, fixed-origin multi-step forecasts, or recursive forecasts.
- Retain the important finding that hybridization does not consistently outperform standalone LSTM.

Background issues, problem formulation and objectives research, when viewed from the clarity of the definition

Good

Further Suggestions:

- The Introduction now presents the forecasting gap, research objective, selected stocks, model comparison, and stock-specific contribution more clearly.
- Strengthen the rationale for selecting coal-sector stocks using direct evidence about Indonesian coal-market exposure, market volatility, and energy-transition risks.
- Distinguish the methodological contribution from the dashboard implementation. A dashboard alone is not a scientific novelty unless usability or decision-support effectiveness is evaluated.
- Add stronger financial-forecasting literature on random-walk benchmarks, market efficiency, forecast evaluation, and hybrid-model limitations.

Research method used, when viewed from the formulation problem and purpose

Enough

Further Suggestions:

- The Methods section has improved substantially through chronological splitting, ARIMA grid search, LSTM configuration, fixed seed, standalone baselines, and expanding-window validation.
- Define the complete forecasting protocol precisely. Explain how ARIMA residuals are generated for LSTM training and testing, and demonstrate that no future actual prices or residuals are used when generating each forecast.
- Clarify whether the 15% LSTM validation subset is the last chronological portion of training data and confirm that training data are not shuffled.
- Re-select ARIMA order separately within every expanding-window fold, rather than applying an order selected using later observations.
- Resolve the ADRO stationarity inconsistency: KPSS indicates that the first-differenced series remains non-stationary, yet ARIMA with (d=1) is retained. Provide additional diagnostics, an alternative specification, or a justified transformation.
- Add ARIMA residual diagnostics, including residual autocorrelation and Ljung–Box testing.
- Provide a versioned and reproducible data snapshot, download date, software/package versions, hardware information, source code, and detailed requirements file. A Yahoo Finance/yfinance data pull alone is not sufficiently reproducible for a reputable journal.

Clarity and comprehensiveness of methods / analysis and synthesis with the results of the discussion, have you answered the problem thoroughly?

No

Further Suggestions:

- Add strong financial forecasting benchmarks, especially a naïve random-walk/no-change benchmark and, where appropriate, drift or seasonal-naïve models. Comparison only with ARIMA and LSTM is insufficient.
- Report statistical tests of forecast differences, such as a Diebold–Mariano test, rather than inferring superiority solely from metric differences.
- Report all metrics, not only mean MAPE, for each expanding-window fold. Include fold-level MAE, RMSE, MAPE, R^2, and dispersion across windows.
- Explain why hybrid models are used for 2025–2045 projections for INDY when standalone LSTM performs substantially better in both final holdout and expanding-window evaluation.

- A 20-year stock-price projection is not supported by a two-year final evaluation and a univariate price-only model. Either remove the 2025–2045 projections or substantially narrow and justify the projection horizon using rolling-origin evidence.
- The ARIMA 95% interval must not be presented as a 95% hybrid-model interval because it excludes uncertainty from LSTM residual correction. Use joint simulation/bootstrap prediction intervals or label these bounds only as ARIMA-component uncertainty.
- Explain how daily forecasts were converted into monthly 2025–2045 projections.
- Keep the current stock-specific interpretation and avoid claims of universal hybrid superiority.

The conclusions in this paper

Enough

Further Suggestions:

- The conclusion appropriately states that hybrid performance is stock-specific and that long-term projections are not investment recommendations.
- However, the journal template requires the Conclusion to be written as **one brief, narrative, non-bulleted paragraph**. Merge the current multi-paragraph conclusion into one paragraph.
- Limit conclusions to the four evaluated stocks, the univariate adjusted-closing-price setup, and the tested periods.
- Do not imply that the dashboard provides validated decision support without a separate usability or decision-quality evaluation.

Have references been written with citation tools?

Yes

Further Suggestions:

- The manuscript uses citation and bibliography fields.
- Replace weakly relevant or low-authority sources with peer-reviewed financial forecasting literature. For example, references related to banking fraud, traffic forecasting, agricultural drought, and general trade balance do not directly support a coal-stock forecasting methodology.
- Replace the OSF source with a peer-reviewed or official source where possible.
- Add an authoritative reference or archived citation for Yahoo Finance/yfinance data and document the retrieval date.
- Verify all DOI metadata and ensure consistent IEEE formatting.

Reviewer 2 — Round 2

Recommendation: Major Revision

Does the title reflect the content, specific and effective?

Yes

Further Suggestions:

- The title now correctly reflects the hybrid forecasting framework and Indonesian coal-sector context.
- Retain “Forecasting” rather than language suggesting investment advice or guaranteed price prediction.

Abstracts and keywords have described the overall content?

Yes

Further Suggestions:

- The abstract is concise, within the template limit, and reports key metrics for all four stocks.
- Clearly identify the forecast horizon and forecast-update mechanism used during the January 2023–December 2024 test period.
- State that long-term projections rely on a univariate historical-price model and are conditional scenarios, not validated long-horizon forecasts.
- The updated keywords are appropriate; consider adding “time-series validation” or “stock-price prediction” if indexing policy permits.

Background issues, problem formulation and objectives research, when viewed from the clarity of the definition

Good

Further Suggestions:

- The study objective and expected contribution are clearer than in Round 1.
- Strengthen the review of prior hybrid ARIMA-LSTM financial studies by comparing their validation designs, forecast horizons, benchmark models, and treatment of structural breaks.
- Explain why a stock-specific model-selection framework is preferable to a single common hybrid architecture, especially because the reported results differ strongly across companies.

Research method used, when viewed from the formulation problem and purpose

Enough

Further Suggestions:

- The chronological holdout and pre-test expanding-window evaluation are appropriate improvements.
- Nevertheless, the candidate-model comparison and final model choice require a more rigorous rolling-origin or nested time-series model-selection procedure.
- Specify whether scaling, ARIMA fitting, residual generation, sequence creation, and LSTM training are repeated entirely within each validation window.
- Explain whether LSTM residual inputs at test time depend only on information available before the forecast origin.
- Justify the use of fixed LSTM hyperparameters without formal hyperparameter tuning; alternatively, tune them inside the training/validation period only.
- Include computational details: package versions, operating environment, CPU/GPU, training time, and inference time if dashboard responsiveness or efficiency is claimed.

Clarity and comprehensiveness of methods / analysis and synthesis with the results of the discussion, have you answered the problem thoroughly?

No

Further Suggestions:

- Add naïve financial benchmarks, including random walk/no-change, before claiming that ARIMA, LSTM, or hybrid models provide useful forecasting value.
- Apply formal forecast-comparison testing and report whether improvements are statistically significant.
- Report directional-accuracy metrics and scaled errors such as MASE or RMSSE, which are more interpretable across stocks than nominal-price metrics alone.
- The expanding-window results show standalone LSTM as the lowest-MAPE model for all four stocks, while the final holdout favors hybrid models for three stocks. Analyze this instability more deeply and do not select a universal hybrid model for projection convenience.
- Remove or sharply qualify the 2025–2045 price projections. Their uncertainty is likely much larger than the ARIMA-only interval presented, and the hybrid correction has no corresponding uncertainty estimate.
- Discuss structural breaks more quantitatively, including potential change points, major coal-price shocks, corporate actions, or regime-specific performance.
- Ensure all figures are readable in black and white, contain independent captions below each figure, and are cited in numerical order according to the journal template.

The conclusions in this paper

Enough

Further Suggestions:

- The conclusion now correctly acknowledges stock-specific performance, the weakness of hybrid models for INDY, and the exploratory nature of projections.
- Revise the Conclusion into one concise narrative paragraph, as required by the journal template.
- State clearly that no model has yet demonstrated reliable 20-year stock-price forecasting capability.
- Emphasize future work involving rolling-origin evaluation, multivariate predictors, structural-break handling, calibrated prediction intervals, and external replication.

Have references been written with citation tools?

Yes

Further Suggestions:

- The reference list appears to be citation-tool managed, but it needs a relevance and quality audit for a reputable journal.
- Prioritize peer-reviewed international finance and time-series forecasting studies over tangential applications from fraud detection, traffic forecasting, drought forecasting, or unrelated economic topics.
- Retain foundational ARIMA, LSTM, and hybrid references, but add stronger recent financial-market forecasting and forecast-evaluation sources.
- Verify DOI resolution, author initials, journal-title italics, article-title quotation marks, and complete IEEE metadata for every reference.